

Summary Chapter 1 Basic concepts of signals and systems



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1 Basic concepts

2 Description and classification of signals

3 Continuous-time signals

4 Unit step signal and unit impulse signal

5 Decomposition of signals

6 Discrete-time signals

7 System

1. Representation of complex numbers

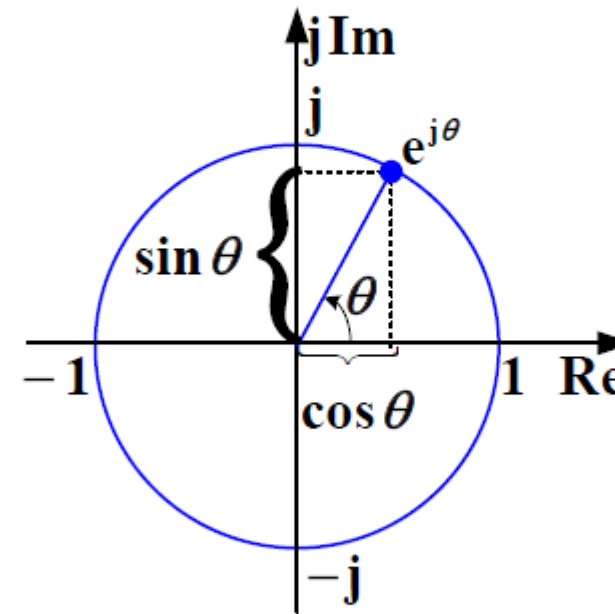
Two forms:

Rectangular form:

$$\cos\theta + j\sin\theta$$

Polar form (complex
exponential form):

$$e^{j\theta}$$



1. Representation of complex numbers

Polar Form of Complex Numbers $Ae^{j\theta}$

Express the following complex numbers in polar form:

$$1+j = \sqrt{2}e^{j\frac{\pi}{4}}$$

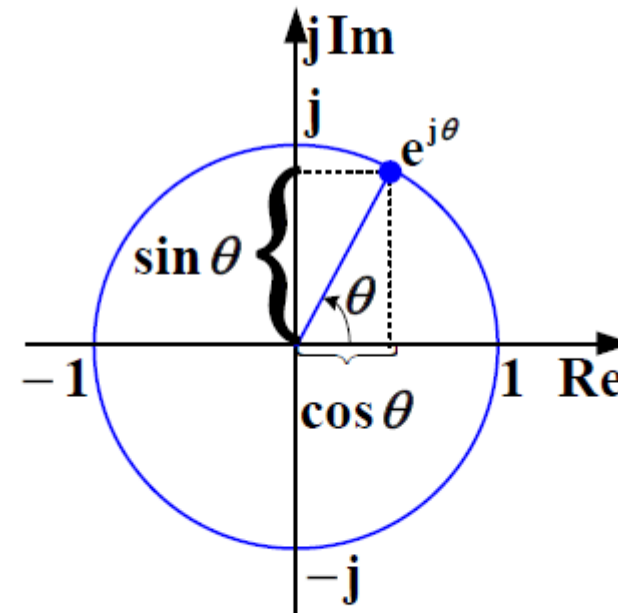
$$-1 = e^{j\pi}$$

$$j = e^{j\frac{\pi}{2}}$$

$$-j = e^{-j\frac{\pi}{2}}$$

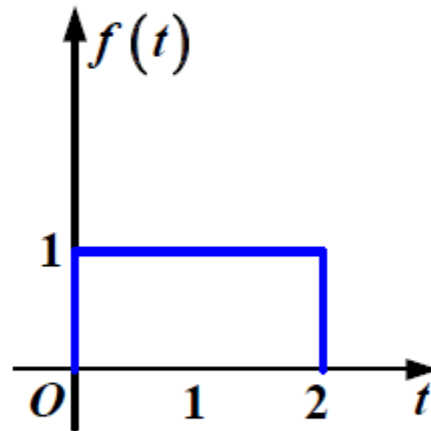
Magnitude?

Phase?



2. Representation and Operations of Continuous-Time Signals

Please get the time function of the waveform



2. Representation and Operations of Continuous-Time Signals

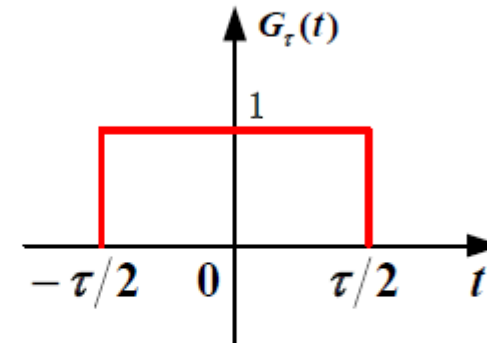
$$\int_{-\infty}^{\infty} \delta(5t) f(t) dt = ?$$

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(5t) f(t) dt &= \int_{-\infty}^{\infty} \frac{1}{5} \delta(t) f(t) dt \\ &= \frac{1}{5} f(0) \end{aligned}$$

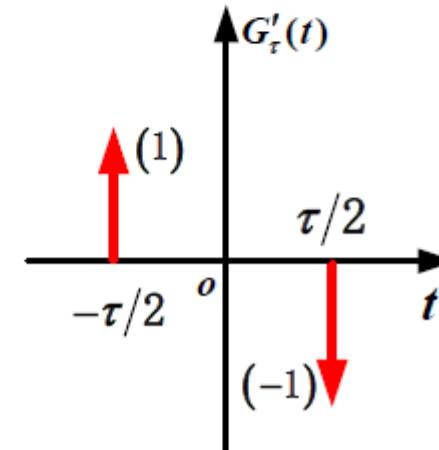
2. Representation and Operations of Continuous-Time Signals

Please plot the waveform of the rectangular function/gate function and its derivative function

$$G_{\tau}(t) = \left[u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right) \right]$$

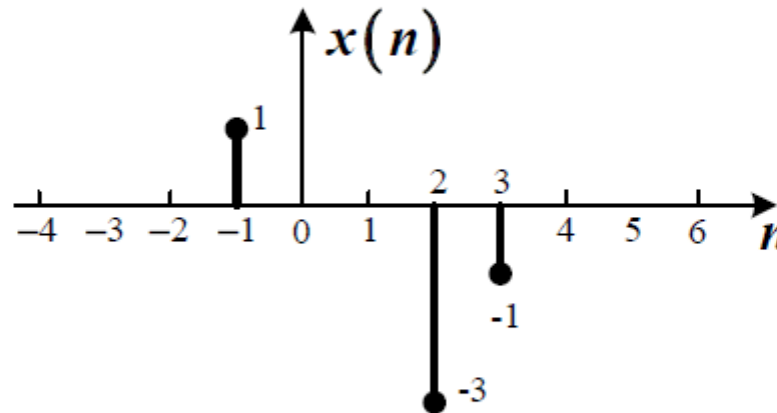


$$G'_{\tau}(t) = \delta\left(t + \frac{\tau}{2}\right) - \delta\left(t - \frac{\tau}{2}\right)$$



3. Representation and Operations of Discrete-Time Signals

Please express $x(n]$ as a linear combination of $\delta(n]$



$$x[n] = \delta[n + 1] - 3\delta[n - 2] - \delta[n - 3]$$

4. Description and Properties of Linear Time-Invariant Systems

A linear time-invariant system has the following:

When the excitation is $x_1(t) = e^{-t}u(t)$, the zero-state response is $y_1(t) = \cos(3t)u(t)$

Find the expression for the zero-state response $y_2(t)$ when the excitation is

$$x_2(t) = -e^{-t}u(t) + e^{-t}\delta(t)$$

And $y_3(t)$

$$x_3(t) = -e^{-t}u(t) + \delta(t)$$

According to the differential and integral properties of linear time-invariant systems:

$$\text{If } x_1(t) \rightarrow y_1(t) \\ \text{then } \frac{dx_1(t)}{dt} \rightarrow \frac{dy_1(t)}{dt}$$

We have $x_1(t) = e^{-t}u(t)$

Differentiate $x_1(t)$

$$\frac{d}{dt}[e^{-t}u(t)] = -e^{-t}u(t) + e^{-t}\delta(t)$$

Note: $e^{-t}\delta(t) = \delta(t)$ because e^{-t} evaluated at $t = 0$ is 1

So

$$x_1'(t) = -e^{-t}u(t) + \delta(t)$$

Since

$$y_1(t) = \cos(3t)u(t)$$

Differentiate

$$y_2(t) = y_1'(t) = -3\sin(3t)u(t) + \cos(3t)\delta(t)$$

Since $\cos(3t)\delta(t) = \delta(t)$ (because $\cos(0) = 1$):

$$y_2(t) = -3\sin(3t)u(t) + \delta(t)$$